

KurBhan Logistics Database Specification

Dokumentasi Skema DDL Database PostgreSQL & Row Level Security (RLS) Supabase

VERSION 2.0 • PRODUCTION READY

Ringkasan Pembaruan Spesifikasi:

- 1. Pembaharuan ENUM jenis armada menjadi mobil_box_kecil, mobil_box_sedang, dan mobil_box_besar.
- 2. Integrasi Master Wilayah Indonesia (Provinsi, Kab/Kota, Kecamatan, Kelurahan) untuk penentuan akurat lokasi & aturan Sameday.
- 3. Skema Master Tarif Logistik & Matriks Multiplier Surcharge Luar Jawa (Transit Laut/Udara).
- 4. Proteksi transaksi RLS: Pembatalan pesanan dibatasi hanya jika status masih pending.

1. Matriks Tarif Dasar & Surcharge Multi-modal

Jenis Armada	Layanan	Cakupan Wilayah	Tarif / Kg	Status Ketersediaan
Mobil Box Kecil	Reguler	Intra / Inter Provinsi	Rp 8.000	Tersedia
Mobil Box Kecil	Cepat	Intra / Inter Provinsi	Rp 10.000	Tersedia
Mobil Box Kecil	Sameday	Satu Provinsi (Intra)	Rp 20.000	Tersedia
Mobil Box Kecil	Sameday	Beda Provinsi (Inter)	-	Tidak Tersedia
Mobil Box Sedang	Reguler	Intra / Inter Provinsi	Rp 14.000	Tersedia
Mobil Box Sedang	Cepat	Intra / Inter Provinsi	Rp 20.000	Tersedia
Mobil Box Sedang	Sameday	Satu Provinsi (Intra)	Rp 32.000	Tersedia
Mobil Box Sedang	Sameday	Beda Provinsi (Inter)	-	Tidak Tersedia
Mobil Box Besar	Reguler	Intra / Inter Provinsi	Rp 20.000	Tersedia
Mobil Box Besar	Cepat	Intra / Inter Provinsi	Rp 30.000	Tersedia
Mobil Box Besar	Sameday	Satu Provinsi (Intra)	Rp 45.000	Tersedia
Mobil Box Besar	Sameday	Beda Provinsi (Inter)	-	Tidak Tersedia

Ketentuan Surcharge Transit Luar Jawa & Estimasi SLA

Jalur Transit	Cakupan Wilayah	Surcharge / Kg	Estimasi Waktu Pengiriman (SLA)
Darat	Pulau Jawa (Intra/Inter)	Rp 0	Sameday: 6–12 Jam Cepat: 1–2 Hari Reguler: 2–4 Hari
Laut	Luar Pulau Jawa (Kapal Kargo)	+ Rp 7.500	Cepat: 4–7 Hari Reguler: 7–14 Hari

JALUR TRANSIT	CAKUPAN WILAYAH	SURCHARGE / KG	ESTIMASI WAKTU PENGIRIMAN (SLA)
Udara	Luar Pulau Jawa (Pesawat Kargo)	+ Rp 22.500	Cepat: 1–2 Hari Reguler: 2–3 Hari

2. Script DDL SQL Lengkap Supabase PostgreSQL

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-- =====
-- 1. EXTENSIONS & CUSTOM ENUMS
-- =====
CREATE EXTENSION IF NOT EXISTS "uuid-ossf";

CREATE TYPE user_role AS ENUM ('customer', 'admin', 'courier');
CREATE TYPE vehicle_type AS ENUM ('mobil_box_kecil', 'mobil_box_sedang', 'mobil_box_besar');
CREATE TYPE service_type AS ENUM ('reguler', 'cepat', 'sameday');
CREATE TYPE transit_type AS ENUM ('darat', 'laut', 'udara');
CREATE TYPE shipment_status AS ENUM ('pending', 'processing', 'in_transit', 'delivered', 'cancelled');

-- =====
-- 2. MASTER WILAYAH INDONESIA
-- =====
CREATE TABLE public.provinces (
    id CHAR(2) PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    is_java_island BOOLEAN DEFAULT FALSE NOT NULL
);

CREATE TABLE public.regencies (
    id CHAR(4) PRIMARY KEY,
    province_id CHAR(2) REFERENCES public.provinces(id),
    name VARCHAR(255) NOT NULL
);

CREATE TABLE public.districts (
    id CHAR(7) PRIMARY KEY,
    regency_id CHAR(4) REFERENCES public.regencies(id),
    name VARCHAR(255) NOT NULL
);

CREATE TABLE public.villages (
    id CHAR(10) PRIMARY KEY,
    district_id CHAR(7) REFERENCES public.districts(id),
    name VARCHAR(255) NOT NULL,
    postal_code VARCHAR(5)
);

-- =====
-- 3. TABEL PROFILES & SHIPPING RATES
-- =====
CREATE TABLE public.profiles (
    id UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,
    email TEXT NOT NULL UNIQUE,
    full_name TEXT,
    phone_number TEXT,
    role user_role DEFAULT 'customer'::user_role NOT NULL,
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    mfa_enabled BOOLEAN DEFAULT FALSE NOT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    updated_at TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

CREATE TABLE public.shipping_rates (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    vehicle_type vehicle_type NOT NULL,
    service_type service_type NOT NULL,
    is_same_province BOOLEAN NOT NULL,
    price_per_kg NUMERIC(12, 2) NOT NULL,
    is_available BOOLEAN DEFAULT TRUE NOT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

-- =====
-- 4. TABEL SHIPMENTS (TRANSAKSI LOGISTIK)
-- =====

CREATE TABLE public.shipments (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    user_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE CASCADE,

    origin_village_id CHAR(10) NOT NULL REFERENCES public.villages(id),
    destination_village_id CHAR(10) NOT NULL REFERENCES public.villages(id),
    origin_address_detail TEXT NOT NULL,
    destination_address_detail TEXT NOT NULL,

    vehicle_type vehicle_type NOT NULL,
    service_type service_type NOT NULL,
    transit_type transit_type NOT NULL,

    actual_weight NUMERIC(10, 2) NOT NULL CHECK (actual_weight > 0),
    length_cm NUMERIC(10, 2) NOT NULL CHECK (length_cm > 0),
    width_cm NUMERIC(10, 2) NOT NULL CHECK (width_cm > 0),
    height_cm NUMERIC(10, 2) NOT NULL CHECK (height_cm > 0),

    volumetric_weight NUMERIC(10, 2) GENERATED ALWAYS AS (
        (length_cm * width_cm * height_cm) / 5000
    ) STORED,
    chargeable_weight NUMERIC(10, 2) GENERATED ALWAYS AS (
        GREATEST(actual_weight, (length_cm * width_cm * height_cm) / 5000)
    ) STORED,

    total_price NUMERIC(12, 2) NOT NULL,
    estimated_delivery_min TIMESTAMPTZ NOT NULL,
    estimated_delivery_max TIMESTAMPTZ NOT NULL,

    status shipment_status DEFAULT 'pending'::shipment_status NOT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    updated_at TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

-- =====
-- 5. TABEL SHIPMENT_TRACKING & CHAT_MESSAGES
-- =====

CREATE TABLE public.shipment_tracking (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    shipment_id UUID NOT NULL REFERENCES public.shipments(id) ON DELETE CASCADE,
    current_location TEXT NOT NULL,
    status shipment_status NOT NULL,
    notes TEXT,

```

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        updated_at TIMESTAMPTZ DEFAULT NOW() NOT NULL
    );

CREATE TABLE public.chat_messages (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    shipment_id UUID REFERENCES public.shipments(id) ON DELETE SET NULL,
    sender_id UUID NOT NULL REFERENCES public.profiles(id) ON DELETE CASCADE,
    receiver_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE,
    message TEXT NOT NULL,
    is_read BOOLEAN DEFAULT FALSE NOT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

-- =====
-- 6. ROW LEVEL SECURITY (RLS) POLICIES
-- =====

ALTER TABLE public.profiles ENABLE ROW LEVEL SECURITY;
ALTER TABLE public.shipments ENABLE ROW LEVEL SECURITY;
ALTER TABLE public.shipment_tracking ENABLE ROW LEVEL SECURITY;
ALTER TABLE public.chat_messages ENABLE ROW LEVEL SECURITY;

CREATE OR REPLACE FUNCTION is_admin(user_id UUID)
RETURNS BOOLEAN AS $$
    SELECT EXISTS (
        SELECT 1 FROM public.profiles
        WHERE id = user_id AND role = 'admin'
    );
$$ LANGUAGE sql SECURITY DEFINER;

-- Aturan RLS Pembatalan: User hanya bisa update status ke cancelled jika masih pending
CREATE POLICY "Users can cancel pending shipments only"
ON public.shipments FOR UPDATE
USING (
    auth.uid() = user_id
    AND status = 'pending'
)
WITH CHECK (
    NEW.status = 'cancelled'
);

```